

Tim Zhang

timzhang.email@gmail.com · 626-418-3666 · Alhambra, CA

UCLA Computer Science undergraduate with experience building software solutions using Python, C++, and machine learning frameworks. Developed projects involving image processing, data analysis, and deep learning while applying software design, debugging, and testing practices. Experienced providing technical support to engineering students and troubleshooting hardware and software issues through UCLA SEASNet. Seeking a Software Engineer Internship to contribute to reliable, user-focused applications.

EDUCATION

B.S. Computer Science, University of California, Los Angeles (UCLA)

Expected May 2028

Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming, Software Design, Foundations of Computer Vision, Introduction to Machine Learning

CS50's Introduction to Programming with Python — Harvard (Certificate Earned)

COMPUTER VISION & ML PROJECTS

CNN-Based Image Classification — CIFAR-10

EC ENGR 149: Foundations of Computer Vision

- Built and trained a CNN image classifier in PyTorch, achieving 65.7% test accuracy on 10-class image recognition.
- Tuned model settings such as batch size and epochs; analyzed deep learning concepts including pooling, padding, and batch normalization.

Robust Geometric Estimation — SIFT, RANSAC & Homography

EC ENGR 149

- Implemented SIFT keypoint matching, a from-scratch RANSAC loop, and direct linear transform (DLT) homography estimation in NumPy/OpenCV to stitch image pairs via inverse warping and bilinear interpolation.

Feature Detection — Edges, Blobs & Corners

EC ENGR 149

- Implemented 2D convolution, Gaussian/median/bilateral filtering, gradient-based edge detection, Laplacian-of-Gaussian blob detection, and Harris corner detection with non-maximum suppression
- Evaluated denoising methods using error metrics and visual analysis to select the best filter for noisy images.

Semantic Matching & Reasoning with Language Models

M146: Introduction to Machine Learning

- Fine-tuned a DistilBERT model for paraphrase detection, improving accuracy from 61.8% to 81.9% using cross-attention between sentence pairs.
- Applied LoRA fine-tuning to Qwen2.5-1.5B for math reasoning, improving GSM8K accuracy from 36% to 49% while training fewer than 0.15% of parameters.

EXPERIENCE

Computer Lab Consultant, UCLA SEASNet

Mar 2026 – Present

- Provide technical support to 100s of engineering students and faculty daily, resolving software, hardware, and network issues to minimize lab downtime; collaborate on lab-wide system improvements.

Translator / Transcriber, TED Translators

Jun 2022 – 2024

- Translated TED Talk scripts between English and Simplified Chinese and produced time-synced subtitles for global accessibility, managing multiple concurrent projects to consistent deadlines.

Mathematics Tutor, Learn To Be

Aug 2023 – Apr 2024

- Tutored K–12 students from underserved communities with individualized lesson plans, communicating progress with families regularly.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, HTML/CSS, SQL

Python Libraries: OpenCV, PyTorch, NumPy, scikit-learn, pandas

Frameworks & Tools: Git, Jupyter / Google Colab, API usage, AI IDEs (Cursor, Claude), Excel, AI CLIs

Soft Skills: Team Communication, Technical troubleshooting communication, Deadline management

Languages (Human): English, Mandarin Chinese